

FEM for TFIM

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1 Introduction

The Ising model is a binary-spin Hamiltonian defined on a graph $G = (V, E)$ where each site $i \in V$ carries a classical spin variable $s_i \in +1, -1$. The most general (pairwise) classical Ising Hamiltonian with external fields is

$$H_{\text{Ising}}(s) = - \sum_{(i,j) \in E} J_{ij} s_i s_j - \sum_{i \in V} h_i s_i,$$

with real couplings J_{ij} and local fields h_i . The equilibrium statistical mechanics at temperature T is defined by the Gibbs measure with partition function

$$Z(\beta) = \sum_{s \in \pm 1^{|V|}} e^{-\beta H_{\text{Ising}}(s)}, \quad \beta \equiv 1/(k_B T),$$

and the thermodynamic free energy $F(\beta) = -\beta^{-1} \ln Z(\beta)$. The ground-state problem is the zero-temperature ($\beta \rightarrow \infty$) minimization of H_{Ising} , i.e. finding $s^* = \arg \min_s H_{\text{Ising}}(s)$. Simple homogeneous ferromagnets (constant positive J_{ij} on a regular lattice and $h_i \equiv 0$) have the trivial minimizing configurations $s_i \equiv +1$ and $s_i \equiv -1$, but the model's expressive power arises when J_{ij} and h_i vary in sign and magnitude.

The graph-theoretic equivalence to combinatorial optimization is obtained by algebraic re-expression of pairwise terms. A "cut" in a graph is a partition of the nodes into two sets. The "size" of the cut is the number of edges between these two sets. Finding the ground state of an antiferromagnetic Ising model ($J_{ij} = -1$) is exactly the same as finding the maximum cut of its graph. Because every edge connecting spins of opposite sign ($\sigma_i \sigma_j = -1$) contributes $-(-1) \times 1 = +1$ to the "size" of the cut (if we define the two sets by spin direction). Maximizing the number of such edges is the Max-Cut problem. Up to an additive constant and an overall

sign, many combinatorial objectives map onto H_{Ising} . A canonical example is Max-Cut on a weighted graph with weights $w_{ij} \geq 0$, whose objective can be written as

$$\text{Cut}(s) = \sum_{(i,j) \in E} w_{ij} \frac{1 - s_i s_j}{2}.$$

Maximizing $\text{Cut}(s)$ is equivalent to minimizing the Ising energy with $J_{ij} = -w_{ij}$ (and $h_i = 0$). Consequently, the ground-state search of a general Ising Hamiltonian is an **NP-hard combinatorial optimization problem** in the worst case (spin-glass instances with mixed-sign J_{ij}), which explains why nontrivial instances require heuristic or exponential-time algorithms.

1.1 Symmetry breaking, nonquilibrium behaviour and phase transition : intro

The Ising model provides the simplest, solvable example of a phase transition. In 1D, it has no phase transition at any finite temperature. But in 2D, Lars Onsager famously found the exact solution in 1944, showing that it undergoes a phase transition at a critical temperature T_c . Below T_c , the system is ordered. Spins are mostly aligned (ferromagnetic), and there is a net magnetization. This is the "ground state-like" phase. However, above T_c , the system is disordered. Thermal fluctuations break the alignment, and the net magnetization is zero. The point is, the ground state is only relevant at absolute zero temperature ($T = 0$). At any finite temperature, the system explores a vast number of configurations, and its average properties change dramatically at T_c . This emergent phenomenon from simple local rules is the real magic.

Phase transitions and symmetry breaking are thermodynamic/collective phenomena distinct from single-instance optimization, but they are mathematically connected to the energy landscape topology. In statistical physics one studies the behavior of macroscopic observables (e.g. magnetization $m = \frac{1}{|V|} \sum_i \langle s_i \rangle$) as functions of T and control parameters. A second-order phase transition occurs when the free energy $F(\beta)$ develops a non-analytic dependence on β , typically accompanied by a spontaneous breaking of a symmetry of H_{Ising} (for example global spin-flip symmetry $s \mapsto -s$ in zero-field ferromagnets). Physically relevant distinctions appear between finite lattices and the thermodynamic limit $|V| \rightarrow \infty$: e.g. the one-dimensional nearest-neighbor classical Ising chain has no finite- T phase transition, whereas the two-dimensional square-lattice Ising model exhibits a nontrivial critical temperature (Onsager solution) and universal scaling near criticality. These collective effects determine the structure of low-energy valleys and barriers, thereby influencing algorithmic performance of thermal or dynamical search methods.

The **transverse-field Ising model (TFIM)** introduces quantum mechanics by replacing classical spins with Pauli operators and adding a *noncommuting* transverse term. The standard TFIM Hamiltonian is

$$\hat{H}_{\text{TFIM}} = - \sum_{(i,j) \in E} J_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z - \sum_{i \in V} h_i \hat{\sigma}_i^x - \Gamma \sum_{i \in V} \hat{\sigma}_i^x,$$

where Γ is the transverse-field strength and $\hat{\sigma}^{x,z}$ are Pauli matrices. Quantum annealing/adiabatic optimization typically interpolates between an easily prepared driver Hamiltonian $\hat{H}_B = -\sum_i \hat{\sigma}_i^x$ and a problem Hamiltonian $\hat{H}_P$ (diagonal in the z basis) via $\hat{H}(s) = (1-s)\hat{H}_B + s\hat{H}_P$ with $s \in [0, 1]$. Adiabatic evolution aims to track the ground state of $\hat{H}(s)$ for

s ramped slowly; the runtime is controlled by the minimum spectral gap $\Delta_{\min}$ along the path and by diabatic excitations. The TFIM exhibits quantum phase transitions as a function of Γ (or s), which in many-body instances correspond to changes in the entanglement structure and closing of the spectral gap; these critical phenomena directly affect annealing performance and the scaling of difficulty.

Finally, the practical point tying physics and optimization is dynamics and landscape complexity. This is because :

1. Frustration (incompatible constraints from mixed-sign J_{ij})
2. Disorder (random fields and even coupling constants like J_{ij}, h_i)
3. Finite-dimensional locality

produce rugged energy landscapes with exponentially many metastable states (spin-glass behavior). Thermal algorithms (simulated annealing) probe these landscapes via stochastic sampling at finite temperature, using temperature schedules to escape local minima; quantum algorithms (quantum annealing, variational dynamics) supplement thermal hopping with coherent tunneling and interference, which can reduce barrier-crossing times in some instances. Nonequilibrium phenomena—quench dynamics, diabatic transitions, and Kibble–Zurek scaling near critical points—are therefore central to understanding whether a given dynamical protocol will reliably find the global minimum or be trapped in a local one.